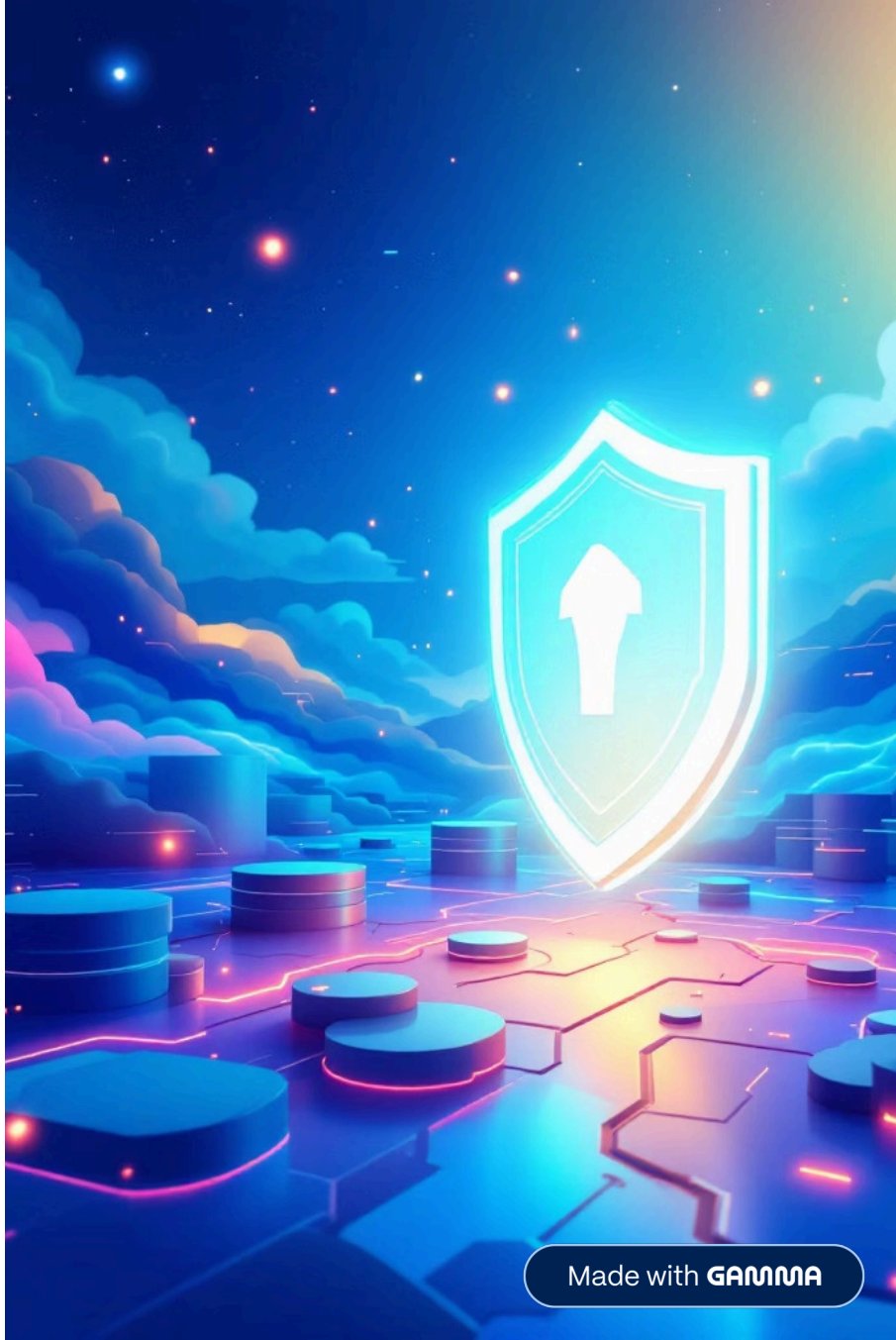
Financial Fraud Detection & Transaction Risk Analysis

Identifying Suspicious Patterns & High-Risk Accounts Through Data-Driven Analysis

Dataset: 50,000 transactions | 495 accounts | Jan 2020 – Dec 2025



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Executive Summary

50K

Transactions Analyzed

6 years of data across ATM, Branch, and Online channels

95.2%

Accounts Use Multiple Devices

471 of 495 accounts show multi-device behavior

14%

High-Amount Transactions

7,021 transactions exceed statistical threshold

609

Devices Linked to Multiple Accounts

Strong signal of shared or cloned credentials

This comprehensive SQL-driven analysis surfaces critical fraud risk signals and delivers actionable intelligence for proactive fraud prevention.

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Critical Fraud Risk Signals

Device Proliferation

609 devices linked to multiple accounts represent the widest attack surface for credential-based fraud

Transaction Outliers

7,021 transactions (14%) exceed $590.69 threshold, requiring automated flagging

Balance-Draining

4,467 transactions consumed >50% of account balance in single event

Geo-Inconsistency

148 accounts (29.9%) have credit transactions from multiple locations

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Top High-Risk Accounts

Account ID	Transactions	Devices Used	Risk Level
AC00363	249	12	Critical
AC00202	241	12	Critical
AC00460	238	12	Critical
AC00362	233	12	High
AC00480	230	10	High

These accounts exhibit maximum transaction frequency combined with extreme multi-device usage patterns, requiring immediate enhanced monitoring.

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Channel & Location Analysis

Channel Distribution

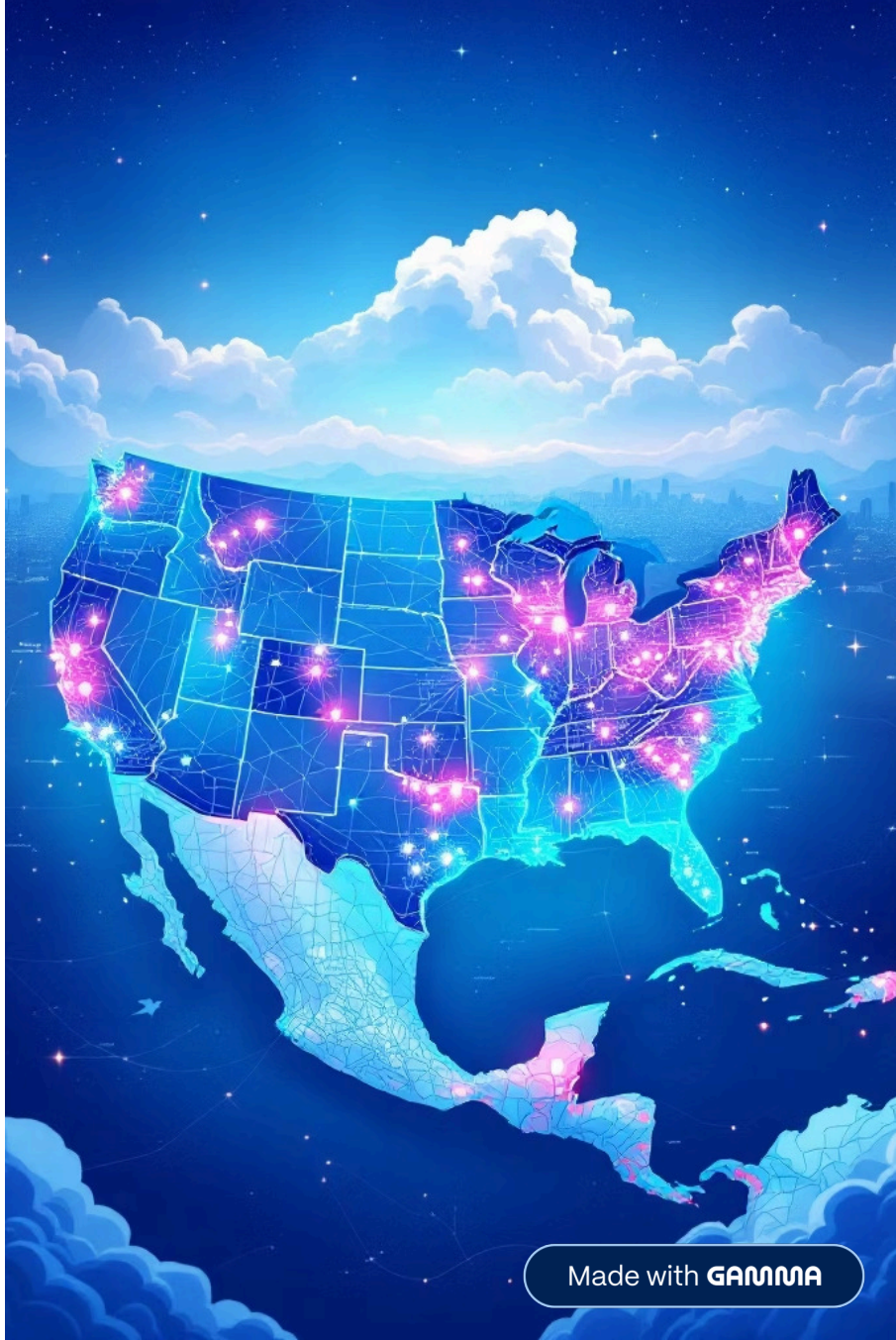
- **Branch:** 17,278 txns ($5.04M) — 34.6%
- **ATM:** 16,552 txns ($5.02M) — 33.1%
- **Online:** 16,170 txns ($4.83M) — 32.3%

All channels show remarkably consistent risk profiles, requiring equal monitoring investment.

Top Locations

1. **Austin:** $453,509
2. **Memphis:** $445,646
3. **Oklahoma City:** $444,560

Top 5 cities account for $2.19M (14.7% of total value) — geographic concentration warrants location-level fraud rules.



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Customer Segment Vulnerabilities

Students — Highest Average Spend

$313.15 average transaction despite lower income profile — potential account misuse requiring monitoring

Age 46-60 — Highest Total Value

$3.82M total spend by age group — high-value target for social engineering and phishing attacks

Age 18-25 — Velocity Risk

$3.01M total despite smaller transaction count — high frequency, lower amounts indicate velocity risk

Retired — Most Vulnerable

$297.94 average — vulnerable demographic requiring elder fraud prevention protocols

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Immediate Action Recommendations

01

Multi-Device Alert System

Deploy device risk scoring: flag devices linked to 5+ accounts; block those linked to 7+ pending verification. Implement device fingerprinting across all channels.

02

Multi-Factor Authentication

Enforce automatic account lock after 3 failed login attempts with 15-minute cooldown and mandatory OTP. Flag transactions after 4th/5th attempt for immediate review.

03

Transaction Velocity Controls

Introduce velocity rule: more than 3 transactions within 30 minutes triggers OTP confirmation. Apply stricter thresholds for Online channel.

Short-Term & Long-Term Strategies

Short-Term (1-3 months)

- **Account-Draining Detection:** Create balance-proportion rules for transactions exceeding 60% of current balance
- **Geo-Inconsistency Flagging:** Build geo-velocity rules for transactions 200+ miles apart within 4 hours
- **Composite Risk Score:** Build fraud risk score incorporating transaction frequency, device count, login attempts, and high-amount flags

Long-Term (3-12 months)

- **Elder Fraud Protection:** Quarterly fraud awareness campaigns for customers aged 55+
- **Student Account Protection:** Introduce soft monthly spending caps with easy override process
- **Annual Fraud Reviews:** Partner with relationship managers for Retired occupation accounts with multi-location patterns

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Implementation Roadmap

1. **Week 1-2: Immediate**

   Deploy device risk scoring API and multi-factor authentication enforcement

2. **Week 3-6: Short-Term**

   Build composite risk score model and geo-velocity alert rules

3. **Month 2-3: Integration**

   Integrate with real-time transaction engine and train fraud analysts

4. **Month 4-12: Long-Term**

   Launch customer protection programs and annual fraud reviews

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Next Steps

Schedule Daily Execution

Run 15 SQL queries daily on refreshed transaction data in PostgreSQL

Build Risk Model

Develop composite fraud risk score incorporating all identified signals

Deploy Protection

Implement device risk scoring and geo-velocity alert rules

This analysis delivered: Complete EDA on 50,000 transactions | 15 SQL fraud detection queries | 6 fraud risk signal categories | Top 5 high-risk accounts profiled | 7 prioritized security recommendations | Power BI dashboard

Target outcome: Reduce real-time fraud losses within 30 days through systematic detection of account-draining and takeover patterns

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